

$$A = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

For Cholesky Decomposition

$$C = (A) \times T^T : (A) \rightarrow 0$$

$$A = LL^T$$

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$l_{11} = \sqrt{a_{11}} = \sqrt{9} = 3$$

$$l_{21} = \frac{15}{3} = 5$$

$$\Rightarrow l_{22} = \sqrt{a_{22} - l_{21}^2} = \sqrt{50 - 25} = 5$$

$$\Rightarrow |l_{22}| = 5$$

$$\Rightarrow L = \begin{bmatrix} 3 & 0 \\ 5 & 5 \end{bmatrix}$$

For $n \times n$ matrix

$$L = \begin{bmatrix} l_{11} & 0 & 0 & \dots & 0 \\ l_{21} & l_{22} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \dots & l_{nn} \end{bmatrix}$$

Diagonal element

$$l_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} l_{ik}^2}$$

For element below diagonal

$$l_{ji} = \frac{a_{ji} - \sum_{k=1}^{i-1} l_{jk} l_{ik}}{l_{ii}}$$